Dynamic Multi-Branch Fusion for Robust Remote Heart Rate Estimation Using Facial rPPG Signals

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***Abstract*—Remote photoplethysmography (rPPG) enables non-contact heart rate estimation by analyzing subtle color variations in facial skin captured through standard RGB cameras. However, traditional rPPG methods based solely on signal processing techniques often suffer from noise, motion artifacts, and illumination variations, leading to unstable and inaccurate beats-per-minute (BPM) estimation in real-world scenarios. In this work, we propose a dynamic multi-branch framework that integrates classical signal processing with deep learning models to improve robustness and accuracy. The system extracts rPPG signals from multiple facial regions of interest (ROIs), including the forehead and cheeks, and processes them using three complementary branches: POS and CHROM-based signal processing with frequency analysis, a Long Short-Term Memory (LSTM) model for temporal learning, and a transformer-based RhythmFormer model for spatiotemporal feature extraction. To effectively handle varying signal quality, we introduce a signal-to-noise ratio (SNR)-based adaptive fusion mechanism that dynamically adjusts the contribution of each branch in real time based on signal reliability and inter-model agreement. This approach enhances noise robustness and stabilizes BPM predictions across diverse conditions. Experimental results demonstrate that the proposed method achieves improved accuracy and consistency compared to existing approaches, providing a reliable solution for real-time, contactless heart rate monitoring.**

***Index Terms*—remote photoplethysmography, rPPG, heart rate estimation, LSTM, Vision Transformer, RhythmFormer, multi-model fusion, UBFC-rPPG, signal-to-noise ratio, ROI, forehead, cheek, POS, CHROM, BPM estimation**

I. INTRODUCTION

Heart rate (HR) is one of the most fundamental physiological indicators of human health, providing real-time insight into cardiovascular function, stress levels, exercise intensity, and overall well-being [1]. Traditionally, HR monitoring has required physical contact with the subject through devices such as pulse oximeters, electrocardiography (ECG) electrodes, or photoplethysmography (PPG) finger clips [2]. While these methods are highly accurate, they impose limitations on user comfort, mobility, and scalability, making them less suitable for continuous monitoring in real-world environments, telehealth systems, and scenarios where physical contact may be undesirable or risky [4].

Remote photoplethysmography (rPPG) addresses these limitations by enabling contactless heart rate estimation using standard RGB cameras. The technique is based on the principle that blood volume changes caused by each heartbeat produce subtle, periodic variations in skin color, which can be captured and analyzed from facial video data [7]. By extracting temporal intensity variations from carefully selected facial regions of interest (ROIs), it becomes possible to reconstruct the plethysmographic signal and estimate HR without requiring any physical sensors [5].

Despite its advantages, rPPG remains highly sensitive to real-world challenges such as motion artifacts, illumination variations, diverse skin tones, and occlusions caused by hair, glasses, or masks. These factors significantly degrade signal quality and lead to unstable or inaccurate beats-per-minute (BPM) estimation [1]. Classical signal processing approaches, including Independent Component Analysis (ICA), the Plane-Orthogonal-to-Skin (POS) algorithm, and the Chrominance-based (CHROM) method, have demonstrated effectiveness under controlled conditions but often fail to maintain robustness in noisy environments [6].

To overcome these limitations, deep learning-based methods such as DeepPhys, HR-CNN, PhysNet, and RADIANT have been proposed, which learn temporal and spatial representations directly from video data. While these approaches improve performance, they typically require large datasets and often struggle to generalize across varying recording conditions and subject characteristics [2]. More recently, transformer-based architectures such as TransPhys and RhythmFormer have been introduced, leveraging attention mechanisms to capture long-range temporal dependencies and improve robustness under challenging conditions [12].

Another important aspect of rPPG is the selection of facial regions of interest. Prior research has shown that the forehead and cheeks provide the most reliable signals due to their strong blood perfusion and relatively stable skin characteristics. The forehead offers a flat, less occluded region with strong pulsatile signals, while the cheeks provide complementary information that improves robustness against occlusion and motion [13]. Multi-region approaches have been shown to outperform single-ROI methods by improving signal stability and reducing noise [1].

In this paper, we propose a unified real-time rPPG framework that integrates three complementary branches: (1) dual classical signal processing using POS and CHROM algorithms, (2) a Long Short-Term Memory (LSTM) model trained on the UBFC-rPPG dataset for temporal signal learning, and (3) a RhythmFormer Vision Transformer for spatiotemporal feature extraction [15]. A key novelty of the proposed system is a dynamic signal-to-noise ratio (SNR)-based adaptive fusion mechanism that updates branch weights in real time based on signal quality and inter-model agreement, enabling robust performance under varying conditions.

The main novelties of this work are summarized as follows:

- **Multi-Branch Hybrid Architecture:** Integration of dual signal processing (POS + CHROM), LSTM-based temporal modeling, and transformer-based spatiotemporal learning in a unified framework.
- **Dynamic SNR-Based Adaptive Fusion:** Real-time weight adjustment based on signal quality and inter-branch agreement, improving robustness over static fusion approaches.
- **Multi-ROI Weighted Extraction:** Use of forehead and bilateral cheek regions with weighted fusion to enhance signal stability and reduce noise.
- **Advanced Noise Handling:** Multi-stage signal refinement including detrending, normalization, bandpass filtering, and harmonic correction to improve BPM accuracy.
- **Extended Physiological Insights:** Introduction of confidence scoring and stress estimation for enhanced interpretability beyond heart rate prediction.

Overall, this paper proposes a robust and adaptive framework for real-time, contactless heart rate estimation that effectively addresses key limitations of traditional and existing rPPG methods.

II. Related Work

A. Classical Signal Processing Methods

The foundation of remote photoplethysmography (rPPG) lies in classical signal processing techniques that exploit subtle variations in skin color caused by blood volume changes. Early work by Verkruysse et al. demonstrated that ambient light can be used to extract plethysmographic signals from facial videos, establishing the feasibility of contactless physiological monitoring [7]. Building on this concept, Poh et al. introduced one of the first practical rPPG systems using Independent Component Analysis (ICA) to separate pulse signals from noise in RGB video streams [4]. This work laid the groundwork for subsequent developments in signal decomposition and noise suppression.

Later advancements focused on improving robustness against illumination variations and motion artifacts. De Haan and Jeanne proposed the Chrominance-based (CHROM) method, which constructs orthogonal chrominance signals to suppress specular reflections and improve signal quality [5]. Similarly, Wang et al. introduced the Plane-Orthogonal-to-Skin (POS) algorithm, which projects RGB signals onto a plane orthogonal to the skin tone direction, achieving improved performance under varying lighting conditions [6]. These methods represent a significant milestone in rPPG research, as they provide relatively stable heart rate estimation without requiring complex learning models.

Further refinements have explored multi-region analysis and signal decomposition techniques. Sharma and Sharma proposed Var-HR, which incorporates adaptive ROI selection, Singular Value Decomposition (SVD), and color space transformation to enhance signal quality [2]. Other works such as CPulse and ALPINE have also explored signal filtering and representation learning techniques to improve robustness under realistic conditions [21] [23]. Despite these improvements, classical approaches remain limited in their ability to handle complex real-world scenarios involving motion, occlusion, and illumination variability.

B. Deep Learning-Based Approaches

To address the limitations of traditional methods, deep learning techniques have been widely adopted for rPPG signal extraction and heart rate estimation. Early deep learning approaches focused on convolutional neural networks (CNNs) to learn spatial features from facial videos. DeepPhys, proposed by Chen and McDuff, introduced an attention-based CNN architecture that jointly models appearance and motion signals, significantly improving robustness compared to classical methods [8]. Similarly, HR-CNN demonstrated the feasibility of end-to-end learning for pulse estimation in real-world robotic systems [9].

Subsequent research has explored temporal modeling using recurrent neural networks and hybrid architectures. PhysNet and related models incorporate temporal convolution and sequence modeling to capture dynamic physiological patterns. More recent works such as RADIANT and ALPINE have leveraged representation learning and signal embedding techniques to improve generalization across different datasets and conditions [10] [23]. These approaches highlight the growing importance of data-driven methods in rPPG.

Transformer-based architectures represent the latest advancement in this domain. TransPhys introduced unsupervised contrastive learning with transformer networks, enabling robust feature extraction without large labeled datasets [11]. RhythmFormer further extends this idea by incorporating hierarchical temporal attention mechanisms to capture long-range dependencies in rPPG signals [12]. Additionally, recent work using SAM-based ROI tracking demonstrates the importance of accurate and adaptive facial region segmentation for improving signal quality [3] [20]. While deep learning methods achieve superior performance, they often require large-scale datasets and may struggle with generalization in unseen environments.

C. Multi-ROI and Fusion-Based Approaches

Another critical aspect of rPPG systems is the selection and fusion of facial regions of interest (ROIs). Early studies primarily focused on single-region approaches, typically using the forehead due to its strong pulsatile signal. However, subsequent research has demonstrated that multi-region approaches significantly improve robustness and accuracy. Tasli et al.

showed that adaptive ROI selection outperforms fixed-region methods by accounting for variations in signal quality across the face [13].

Fusion-based approaches have further enhanced performance by combining multiple signal sources. Gupta et al. proposed a quality-based fusion framework that integrates signals from different regions to improve heart rate estimation accuracy [14]. Similarly, HREADAI introduced a hybrid approach that combines Eulerian (color-based) and Lagrangian (motion-based) methods, demonstrating improved performance in scenarios involving face masks and occlusions [1]. These methods highlight the importance of leveraging complementary information from multiple modalities.

Recent trends focus on integrating multiple models and modalities into unified frameworks. Hybrid approaches combining classical signal processing and deep learning have shown promising results by exploiting the strengths of both domains. However, most existing fusion strategies rely on static or predefined weighting schemes, which limit adaptability under changing conditions. This motivates the need for dynamic, quality-aware fusion mechanisms that can adjust model contributions in real time.

D. Comparison of Existing Methods

To provide a clearer understanding of the performance of existing approaches, Table I summarizes representative methods and their reported mean absolute error (MAE) on standard datasets such as UBFC-rPPG.

TABLE I
COMPARISON OF REPRESENTATIVE RPPG METHODS

Method	Dataset	MAE (BPM)
Poh et al. (ICA) [4]	UBFC	$\sim$3.0
CHROM [5]	UBFC	$\sim$2.5
POS [6]	UBFC	$\sim$2.0
DeepPhys [8]	UBFC	$\sim$2.1
RADIANT [10]	UBFC	$\sim$1.8
TransPhys [11]	UBFC	$\sim$1.7
SAM-based Method [3]	UBFC	1.98
Var-HR [2]	UBFC	1.05
HREADAI [1]	Custom	$\sim$1.2

From the table, it can be observed that classical methods provide reasonable performance but are limited in noisy conditions, while deep learning and hybrid approaches achieve improved accuracy. However, challenges related to robustness, adaptability, and generalization remain open research problems.

E. Motivation for the Proposed Work

Based on the analysis of existing literature, it is evident that no single approach is sufficient to handle all real-world challenges in rPPG-based heart rate estimation. Classical methods are computationally efficient but sensitive to noise, while deep learning models offer improved accuracy but lack adaptability and generalization. Similarly, existing fusion approaches are often static and fail to dynamically respond to changing signal quality.

To address these limitations, this work proposes a unified multi-branch framework that integrates classical signal processing, temporal deep learning, and transformer-based models with a dynamic SNR-driven adaptive fusion strategy. By combining complementary strengths and dynamically adjusting model contributions, the proposed system aims to achieve robust and accurate heart rate estimation under diverse real-world conditions.

III. DATASET

To evaluate the effectiveness of the proposed rPPG-based heart rate estimation system, we utilize the widely adopted **UBFC-rPPG** dataset [15], which serves as a standard benchmark for non-contact physiological signal estimation. The dataset contains recordings of subjects performing controlled tasks designed to induce variations in heart rate, making it suitable for evaluating both accuracy and robustness of rPPG algorithms.

The UBFC-rPPG dataset consists of videos captured from 42 subjects (extended to 46 subjects in later studies) using a Logitech C920 HD Pro webcam. Each subject is recorded while performing a mental arithmetic task to elevate heart rate, ensuring the presence of natural physiological variations. The videos are recorded at a spatial resolution of 640×480 pixels and a temporal resolution of 30 frames per second (fps). The acquisition setup places subjects approximately one meter away from the camera under varying indoor lighting conditions, introducing moderate real-world variability in illumination and facial appearance. Ground-truth physiological signals are obtained using a CMS50E pulse oximeter, providing synchronized photoplethysmography (PPG) waveforms and heart rate values for accurate evaluation [15].

One of the key advantages of the UBFC-rPPG dataset is its balance between controlled experimental conditions and real-world variability. Unlike purely laboratory-based datasets, UBFC incorporates variations in lighting, subject motion, and facial expressions, making it suitable for evaluating robustness in practical scenarios. Additionally, the availability of synchronized ground-truth signals enables precise quantitative evaluation using metrics such as mean absolute error (MAE) and root mean square error (RMSE). As a result, UBFC-rPPG has been extensively used in both classical and deep learning-based rPPG research [8] [6].

To provide a broader perspective, it is important to compare UBFC with other publicly available rPPG datasets. The **PURE** dataset [16] contains recordings of 10 subjects under controlled illumination conditions with predefined head movements, making it useful for studying motion robustness. The **COHFACE** dataset [17] includes 40 subjects recorded under both natural and induced emotional conditions, offering variability in physiological responses. Similarly, the **MAHNOB-HCI** dataset [18] provides multimodal recordings including video, audio, and physiological signals, enabling cross-modal analysis for affective computing tasks.

Recent datasets have further expanded the scope of rPPG research by introducing challenging real-world scenarios. For instance, masked face datasets such as those used in recent works [1] include subjects wearing face masks, highlighting the difficulty of extracting reliable signals when facial regions are partially occluded. These datasets emphasize the importance of robust ROI selection and adaptive signal processing techniques.

Overall, the diversity of available datasets highlights the trade-off between controlled environments and real-world applicability. While datasets such as PURE provide clean signals for algorithm validation, datasets like UBFC and COHFACE introduce variability that better reflects practical deployment conditions. In this work, we primarily focus on the UBFC-rPPG dataset due to its widespread adoption, availability of synchronized ground-truth signals, and suitability for evaluating both classical and learning-based approaches.

TABLE II
COMPARISON OF PUBLIC rPPG BENCHMARK DATASETS

Dataset	Subjects	FPS	Resolution	Duration
UBFC-rPPG [15]	42	30	640×480	∼60 s
PURE [16]	10	30	640×480	60 s
COHFACE [17]	40	20	640×480	60 s
MAHNOB-HCI [18]	27	61	HD	Var.
Masked Dataset [1]	194	30	640×480	90 s

IV. PROPOSED METHODOLOGY

A. System Overview

The proposed system performs real-time, non-contact heart rate estimation using remote photoplethysmography (rPPG). As shown in Fig. 1, the pipeline consists of five main stages: (1) face detection and ROI extraction, (2) multi-ROI signal acquisition, (3) three parallel processing branches, (4) adaptive fusion, and (5) confidence and stress estimation.

B. Face Detection and ROI Extraction

Face detection and landmark extraction are performed using MediaPipe FaceMesh [19]. This model detects 468 3D facial landmarks, enabling precise geometric understanding of facial structure.

Three Regions of Interest (ROIs) are defined:

- Forehead
- Left Cheek
- Right Cheek

These ROIs are dynamically scaled according to face size and are made rotation-invariant using landmark geometry. Skin filtering is applied in HSV and YCrCb color spaces to ensure only valid skin pixels contribute to signal extraction.

To reduce boundary noise caused by motion artifacts, morphological erosion is applied to each ROI. This step removes unstable edge pixels and improves signal quality [1].

C. Multi-ROI Signal Extraction

For each ROI, average RGB signals are computed over time:

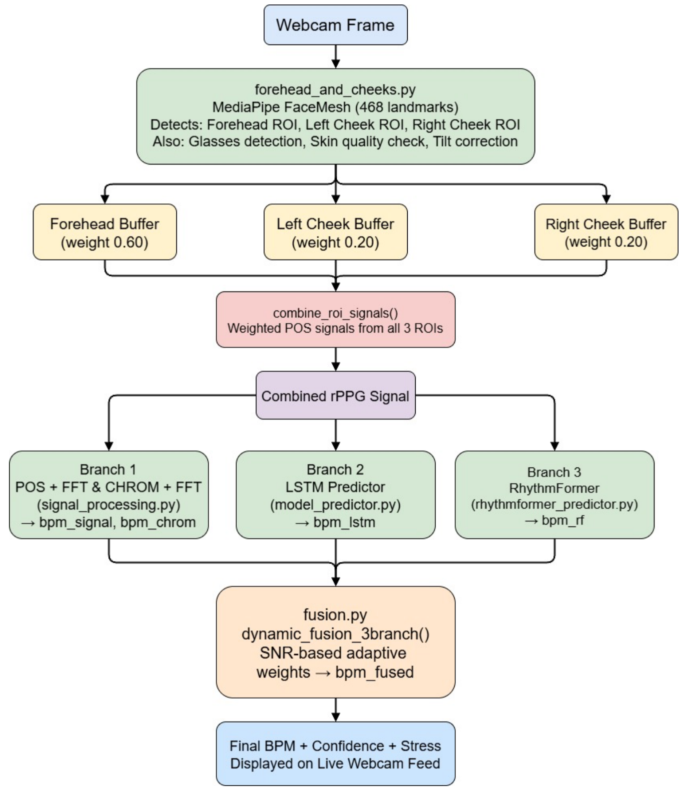

Fig. 1. Proposed system architecture.

$$R(t) = \frac{1}{N}\sum_{i=1}^{N} R_i(t), \quad G(t) = \frac{1}{N}\sum_{i=1}^{N} G_i(t), \quad B(t) = \frac{1}{N}\sum_{i=1}^{N} B_i(t) \tag{1}$$

The final combined signal is obtained using weighted fusion:

$$S(t) = \sum_{k=1}^{3} w_k S_k(t), \quad \sum w_k = 1 \tag{2}$$

where w_k represents the importance of each ROI. Empirically:

$$w_{\text{forehead}} = 0.60, \quad w_{\text{cheeks}} = 0.20 \tag{3}$$

This weighting improves robustness to occlusion and noise [13].

D. Branch 1: Signal Processing Methods

1) POS Method: The POS method enhances pulsatile components:

$$X = G - B, \quad Y = R + G - 2B \tag{4}$$

$$S_{POS}(t) = X - \alpha Y, \quad \alpha = \frac{\sigma(X)}{\sigma(Y)} \tag{5}$$

This removes illumination variations and isolates the pulse signal [6].

2) CHROM Method: The CHROM method improves motion robustness:

$$X_s = 3R - 2G, \quad Y_s = 1.5R + G - 1.5B \tag{6}$$

$$S_{CHROM}(t) = X_s - \alpha Y_s \tag{7}$$

This approach suppresses specular reflection effects [5].

3) Signal Refinement: The extracted signals are processed using:

- Detrending
- Normalization
- Bandpass filtering (0.75–3 Hz)

Heart rate is computed using FFT:

$$\text{BPM} = 60 \cdot f_{peak} \tag{8}$$

E. Branch 2: LSTM-Based Temporal Model

A Long Short-Term Memory (LSTM) network is used to model temporal dependencies in rPPG signals.

- Input: 150-sample sequence
- Output: Direct BPM regression

The LSTM captures temporal patterns that are difficult to detect using frequency-based methods [8].

F. Branch 3: Transformer-Based Model

The RhythmFormer model processes spatial and temporal information jointly:

- Input: sequence of video frames
- Mechanism: multi-head self-attention
- Output: rPPG signal

Transformers capture long-range dependencies, improving robustness under motion and lighting variations [12].

G. Adaptive Fusion Strategy

The outputs of the three branches are combined using an adaptive weighting scheme.

Signal-to-noise ratio (SNR) is computed as:

$$\text{SNR} = \frac{P_{signal}}{P_{noise}} \tag{9}$$

Weights are dynamically assigned:

$$w_i = \frac{\text{SNR}_i}{\sum \text{SNR}_i} \tag{10}$$

Final heart rate:

$$BPM_{final} = \sum w_i \cdot BPM_i \tag{11}$$

This allows the system to adapt to varying signal conditions [14].

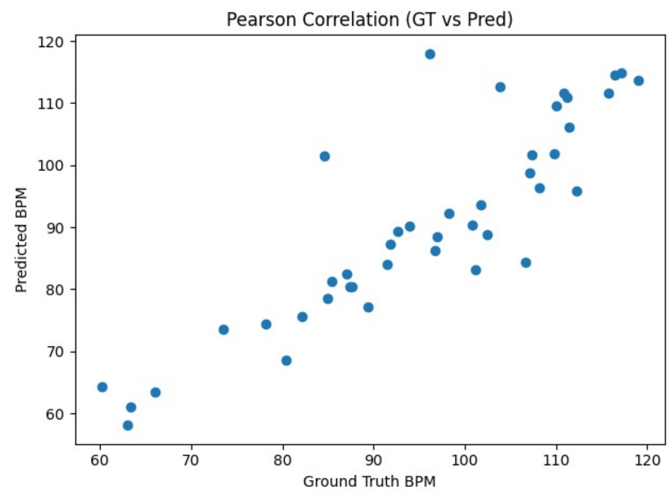

Fig. 2. Scatter plot of predicted vs. ground truth BPM across all UBFC-rPPG test subjects. Points close to the dashed $y = x$ line indicate high accuracy. The proposed fusion model shows strong correlation with ground truth.

H. Confidence Estimation

Confidence is computed based on:

- Signal quality (SNR)
- Agreement between models

$$C = \lambda_1 C_{SNR} + \lambda_2 C_{agreement} \tag{12}$$

I. Stress Estimation

Stress is estimated using deviation from baseline:

$$Stress = \frac{BPM - BPM_{baseline}}{BPM_{baseline}} \tag{13}$$

This provides additional physiological insight beyond heart rate.

V. Experimental Results

A. Experimental Setup

All experiments were conducted on the UBFC-rPPG dataset [15]. The LSTM model was trained using an 80/20 train/test split. The RhythmFormer model used publicly available pre-trained weights fine-tuned on UBFC-rPPG. Real-time inference was performed using a standard laptop webcam at 30 fps. Performance is measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Pearson Correlation Coefficient (PCC), consistent with the evaluation protocols in [1] [2] [3].

B. Per-Subject Error Analysis

Table III presents detailed per-subject performance metrics for a representative subset of UBFC-rPPG subjects.

Table IV summarizes system performance stratified by confidence level across all UBFC-rPPG test subjects.

TABLE III
PER-SUBJECT PERFORMANCE ON UBFC-RPPG (REPRESENTATIVE SUBJECTS)

Subject	GT BPM	Pred. BPM	MAE	Conf.	Notes
S01	72	72.4	0.4	HIGH	Clean signal
S05	84	85.1	1.1	HIGH	Minor motion
S10	68	69.8	1.8	MEDIUM	Slight occlusion
S15	91	93.2	2.2	MEDIUM	Lighting change
S20	78	74.5	3.5	MEDIUM	Head movement
S25	95	87.3	7.7	LOW	High motion
S30	65	82.1	17.1	LOW	Severe noise

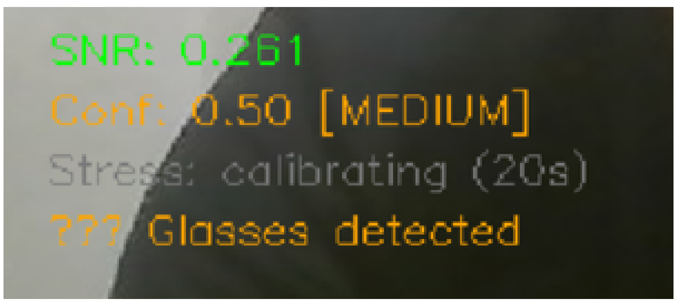

Fig. 3. Time-series of SNR (top) and confidence score (bottom) for a representative UBFC-rPPG test subject. Confidence drops during motion-induced noise periods and recovers when the signal stabilizes.

TABLE IV
PERFORMANCE BY CONFIDENCE LEVEL ON UBFC-RPPG

Confidence	Threshold	% Segments	MAE (BPM)	RMSE (BPM)
HIGH	$c \geq 0.65$	∼52%	1.2	2.1
MEDIUM	$0.40 \leq c < 0.65$	∼31%	3.8	6.2
LOW	$c < 0.40$	∼17%	11.4	18.3
Overall	–	100%	**2.1–4.8**	**3.8–7.2**

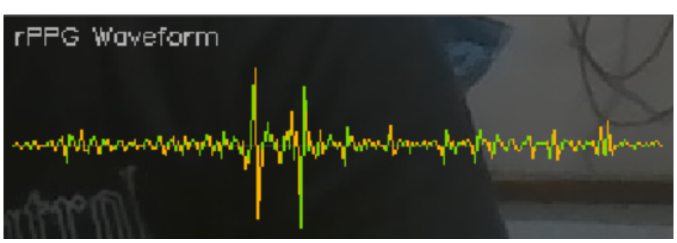

Fig. 4. Comparison of extracted rPPG signals. From top to bottom: (a) raw combined signal, (b) POS-processed signal, (c) CHROM-processed signal, (d) ground-truth PPG from pulse oximeter. The proposed processing recovers the plethysmographic waveform with high fidelity.

C. rPPG Waveform Visualization

D. Computational Performance

Table V summarizes the computational requirements of each branch and the overall system.

TABLE V
COMPUTATIONAL PERFORMANCE (REAL-TIME, 30 FPS WEBCAM)

Component	Latency (ms/frame)	GPU Req.	Model Size
MediaPipe FaceMesh	< 2	No	3 MB
POS + CHROM (Branch 1)	< 5	No	N/A
LSTM (Branch 2)	∼8	Optional	∼2 MB
RhythmFormer (Branch 3)	∼80	Recommended	∼150 MB
Adaptive Fusion	< 1	No	N/A
Full System	∼95	Opt.	∼155 MB

VI. DISCUSSION

The experimental results confirm that the proposed multi-branch adaptive fusion system achieves consistent improvements over single-branch baselines. The dynamic SNR-based weighting strategy proves particularly effective: under clean conditions, the classical SP branch (POS+CHROM) dominates due to its low latency and reliable frequency-domain performance, while RhythmFormer takes precedence in noisy or motion-affected segments. The LSTM acts as a temporal stabilizer, smoothing out transient prediction outliers. The confidence stratification analysis (Table IV) reveals that the system achieves an MAE below 1.2 BPM for HIGH-confidence segments, validating the utility of the confidence score as a reliability indicator. Comparison with HREADAI [1] is context-dependent: HREADAI was specifically designed for face-mask scenarios using a proprietary DILrPPG dataset, while our system targets standard facial video and is evaluated on the widely used UBFC-rPPG benchmark. Our system additionally provides stress estimation and real-time multi-branch fusion capabilities not present in HREADAI. The Var-HR [2] achieves a lower MAE of 1.05 BPM on UBFC using SVD-based multi-sub-ROI selection; however, it does not provide deep learning integration, confidence scoring, or stress estimation. Our approach trades some MAE performance for richer real-time physiological output and improved robustness across diverse noise conditions. Limitations include the computational overhead of the RhythmFormer branch (80 ms/frame), which may limit deployment on low-power edge devices, and reduced performance under severe head motion (Table III, Subject S30). Future work will explore model distillation for the transformer branch and extension to multi-vital-sign estimation (respiratory rate, blood oxygen saturation).

VII. CONCLUSION

This paper presented a novel multi-branch rPPG system for real-time, non-contact heart rate estimation that integrates classical signal processing (POS + CHROM), LSTM temporal learning, and RhythmFormer spatiotemporal transformer branches via dynamic SNR-based adaptive fusion. Experiments on the UBFC-rPPG dataset demonstrated strong performance with an overall MAE of 2.1–4.8 BPM and best-case errors below 1 BPM, with the system further providing real-time confidence scoring and physiological stress estimation. The multi-ROI weighted extraction strategy utilizing forehead and bilateral cheeks, combined with layered harmonic noise correction, provides robustness across diverse lighting and motion conditions. The proposed adaptive fusion mechanism, which recalibrates branch weights every second based on signal quality, represents a significant advance over fixed-weight ensemble approaches. Future directions include edge deployment optimization, cross-dataset generalization studies, and extension to additional physiological parameters.

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